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Assignment 1

Fundamentals of Statistical Learning Theory

Statistical Learning Theory
2026

Question 1

This question considers supervised learning in a broad context. Describe a broad framework for supervised learning by discussing the following points:

- a) The probabilistic approach to supervised learning.
- b) The role of loss functions in the formulation of the objective of a supervised problem. Discuss how different loss functions relate to different supervised models.
- c) Fitting models via empirical risk minimization and the idea behind regularization. Explain the effect of regularization on the empirical risk and the model's generalization capacity.
- d) The importance of measuring the generalization performance of an estimated model, and what different performance estimators explain about the model or learning algorithm.
- e) The relevance of the bias-variance trade-off and curse of dimensionality in statistical learning, with reference to both regression and classification.
- f) Model assessment and selection techniques.

Note: You are strongly encouraged to provide code to illustrate the concepts. The code can be based on real data or simulated examples, where the latter is probably better for the purpose of this question. The content of this question is closely related to Chapter 1 of the course. You are expected to provide additional theoretical or practical examples. One approach would be to generate data from a known distribution and known model, and then illustrate the concepts by numerically computing them.

Question 2

This question entails the analysis of a practical data set. The main objective is to apply a statistical learning method to data and to practically investigate fundamental concepts such as model complexity, model selection, and model assessment.

- a) The first step is to identify a data set you would like to analyze. There are no restrictions and you can download any data set you wish to analyze. You may also scrape/collect data yourself. A good starting point is either this repository, Kaggle, Hugging Face or Zindi. You are allowed to use data from the ESL textbook, data that you used in your Honours project, or simulated data. Credits will be given to the novelty and complexity of the data or application (e.g. scraping data yourself or using a novel data set is better than using a popular open-source data set).

- b) Identify one or several statistical learning methods to apply to the data set identified in (a). You should provide a brief overview of the methods you have identified. This may be a class of models (e.g. neural networks) or different model classes (e.g. linear models, ensembles, kernel methods). There is no restriction on the models other than it must form part of statistical/machine learning research.
- c) Apply the method or methods to the selected data, and summarize the results of your analysis. Clearly distinguish between model training, selection and assessment, highlighting for each of them important considerations in your application.
- d) Clearly report on your findings. Discuss the effect of model selection and assessment on the findings, highlight the strong aspects of your approach and the potential limitations.

Note: It is required to submit code for this question. Also provide a link to the data or the actual data set (if collected yourself). The code should contain all scripts used to collect, import and preprocess the data, as well as all model building and assessment steps. You may also provide a GitHub link to the code and reference it in your report.

Guidelines:

- Define one or several research objectives for the question. E.g. investigating the effect of model complexity on predictive performance; comparing different model selection approaches; etc.
- Broadly follow the structure: introduction, description of data and objective, summary of learning algorithms used, method, results, and conclusion. You can modify this structure to suit your report, but each of these points must be included somewhere.
- Provide sufficient information and visualizations of the data to give the reader a clear understanding of the application.
- Ensure that figures are saved to a high-resolution format, save models to files, and set random states for reproducibility.
- Provide all references to related methods and ideas where applicable.